

Prob: Use the method of Frobenius to find solution of PE, $x^2 y'' + (x^2 - 3x)y' + 3y = 0$

$\Rightarrow$ Do it home

Bessel's Equation and Bessel's function

The DE, $x^2 y'' + x y' + (x^2 - n^2)y = 0$ — (1), where n is a parameter is called Bessel's DE equ. of order n .

Any solution of Bessel equation of order n is called a Bessel function order n .

If $n=0$ then (1)

Prob: Find the solution of the Bessel's equation of order n .

Solⁿ:

We know, the Bessel's equation of order n is,

$$x^2 y'' + x y' + (x^2 - n^2)y = 0 \text{ — (1)}$$

Here $x=0$ is a regular singular point

Let $y = \sum_{n=0}^{\infty} a_n x^{n+\alpha}$ — (2) be a solⁿ of (1)

$$y' = \sum_{n=0}^{\infty} a_n (n+\alpha) x^{n+\alpha-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+\alpha) (n+\alpha-1) x^{n+\alpha-2}$$

Substituting these in Eq (1) and Simplifying,

$$a_0 \alpha(\alpha-1) x^{\alpha} + a_1 (\alpha+1) \alpha x^{\alpha+1} + \sum_{n=2}^{\infty} a_n (\alpha+n) (\alpha+n-1) x^{\alpha+n} +$$

$$a_0 \alpha x^{\alpha} + a_1 (\alpha+1) x^{\alpha+1} + \sum_{n=2}^{\infty} a_n (\alpha+n) x^{\alpha+n} +$$

$$\sum_{n=2}^{\infty} a_{n-2} x^{\alpha+n} = \tilde{n} a_0 x^{\alpha} + \tilde{n} a_1 x^{\alpha+1} - \tilde{n} \sum_{n=2}^{\infty} a_n x^{\alpha+n}$$

equating to zero the co-efficient of $x^{\alpha+1}$, we get

$$a_0 (\alpha^2 - \tilde{n}) = 0 ; a_0 \neq 0$$

$$\alpha^2 = \tilde{n} \therefore \alpha = \pm \sqrt{\tilde{n}}$$

Also, equating to zero the co-efficient of $x^{\alpha+1}$, we get

$$a_1 \{ (\alpha+1)^2 - \tilde{n} \} = 0, \quad a_1 = 0,$$

$$(1+2) \sqrt{\tilde{n}} \neq 0$$

Equating to zero the co-efficient of x^{n+r} we get recurrence formula,

$$a_n[(x+r)^2 - n^2] + a_{n-2} = 0$$

$$\therefore a_n = - \frac{1}{(x+r)^2 - n^2} a_{n-2} ; n > 2 \quad (2)$$

Putting $x = n$ in eq (2)

$$a_n = - \frac{1}{(nn)^2 - n^2} a_{n-2}$$

$$n = 2, 3, 4, \dots$$

$$a_2 = - \frac{1}{2 \cdot 2^2 (n+1)} a_0 ; a_3 = a_5 = a_7 = \dots = 0 \text{ Since } a_1 = 0$$

$$\therefore a_4 = \frac{1}{2 \cdot 4 \cdot 2^2 (n+1)(n+2)} a_0$$

with these values of a 's we get from (2)

$$y = a_0 x^n \left[1 - \frac{x^2}{2 \cdot 2^2 (n+1)} + \frac{x^4}{2 \cdot 2 \cdot 2^2 (n+1)(n+2)} + \dots + (-1)^n \frac{x^{2n}}{2^n (n!)^2} \right]$$

Putting,

$\alpha = -n$ in Eq(2) we get the corresponding

Solⁿ,

$$y_2 = a_0 x^n \left[1 - \frac{x^2}{2 \cdot 2(-n)} + \frac{x^4}{2^2 \cdot 2 \cdot 2^2(-n)(-n-1)} + \dots \right]$$

G.S of Eq(1) is $y = C_1 y_1 + C_2 y_2$.

Bessels function:

The Bessel's DE of order n is $x^2 y'' + xy' + (x^2 - n^2)y = 0$

A solution of this in the form of a power series of ascending power of x is,

$$y = a_0 x^n \left[1 - \frac{x^2}{2 \cdot 2(n+1)} + \frac{x^4}{2 \cdot 4 \cdot 2(n+1)(n+2)} + \dots \right]$$

$$(-1)^n \frac{x^{2n}}{2^n n! 2^n (n+1)(n+2) \dots}$$

where a_0 is an arbitrary constant

If $a_0 = \frac{1}{2^n n!}$ then eq (1) is defined by $J_n(x)$ which is the Bessel's function of order n .

$$J_n(x) = \frac{1}{2^n n!} x^n \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{2^n r! 2^n (n!) (n+r) \dots (n+r)} \quad \text{---}$$

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! (n+r)!} \left(\frac{x}{2}\right)^{n+2r}$$

Pro. Perties :

① Prove that $\frac{d}{dx}[J_0(x)] = -J_1(x)$

Solⁿ:

we know, that,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! (n+r)!} \left(\frac{x}{2}\right)^{n+2r} \quad \text{---}$$

$$\therefore J_0(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! r!} \left(\frac{x}{2}\right)^{2r}$$

$$\frac{d}{dx}[J_0(x)] = \frac{d}{dx} \sum_{r=0}^{\infty} \frac{(-1)^r}{r! r!} \left(\frac{x}{2}\right)^{2r}$$

$$= \sum_{r=1}^{\infty} \frac{(-1)^r}{r! (r-1)!} \left(\frac{x}{2}\right)^{2r-1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(n+1)n!} \left(\frac{x}{2}\right)^{n+1} \left[\text{Replacing } n \text{ by } n+1 \right]$$

$$= - \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1)!} \left(\frac{x}{2}\right)^{n+1}$$

$$= - J_1(x)$$

$$\therefore \frac{d}{dx} [J_0(x)] = -J_1(x)$$

✱

Proof and Property-2

$$\textcircled{2} J_{-n}(x) = (-1)^n J_n(x)$$

Solution. We have by definition

$$J_0(x) = \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{2r}$$

Differentiating it w.r.t. x , we get

$$\frac{d}{dx} J_0(x)$$

$$= \sum_{r=1}^{\infty} (-1)^r \frac{1}{(r)! (r-1)!} \left(\frac{x}{2}\right)^{2r-1}$$

$$= \sum_{j=0}^{\infty} (-1)^{j+1} \frac{1}{(j+1)! (j)!} \left(\frac{x}{2}\right)^{2j+1}$$

where $j = r - 1$

$$= - \sum_{j=0}^{\infty} (-1)^j \frac{1}{(j)! (j+1)!} \left(\frac{x}{2}\right)^{2j+1} = -J_1(x).$$

8.3. When n is positive integer,

$$J_{-n}(x) = (-1)^n J_n(x).$$

Proof. We have

$$J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{(r)! \Gamma(r-n+1)} \left(\frac{x}{2}\right)^{-n+2r}$$

In this series Γ of a negative number occurs in denominator, for terms when $r=0, 1, 2, \dots, (n-1)$ and Γ of a negative number being infinite, all these terms are zero.

Thus terms are obtained from $r=n$ onwards

$$\text{or } J_{-n}(x) = \sum_{r=n}^{\infty} (-1)^r \frac{1}{(r)! \Gamma(r-n+1)} \left(\frac{x}{2}\right)^{-n+2r}$$

Now take $r=n+j$, so that when $r=n, j=0$ and

$$J_{-n}(x) = \sum_{j=0}^{\infty} (-1)^{j+n} \frac{1}{(n+j)! \Gamma(n+j-n+1)} \left(\frac{x}{2}\right)^{-n+2n+2j}$$

$$= \sum_{j=0}^{\infty} (-1)^{j+n} \frac{1}{(n+j)! \Gamma(j+1)} \left(\frac{x}{2}\right)^{n+2j}$$

$$= (-1)^n \sum_{j=0}^{\infty} (-1)^j \frac{1}{(n+j+1)! (j)!} \left(\frac{x}{2}\right)^{n+2j}$$

as $(n+j)! = \Gamma(n+j+1)$ and $\Gamma(j+1) = (j)!$

$$= (-1)^n J_n(x).$$

Note. Because of this relation in $J_n(x)$ and $J_{-n}(x)$, the two functions are not independent when n is an integer. Therefore in this case a second solution of Bessel's equation is to be found.

Ex. State the Bessel's differential equation and prove that its most general solution is $y = AJ_n(x) + BJ_{-n}(x)$ where A and B are arbitrary constants. [Agra 1973]

Differentiating w.r.t. x , we get

$$\begin{aligned}
 2J'_n &= \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= \sum_{r=0}^{\infty} \frac{(-1)^r \{(n+r)+r\}}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &\quad \text{taking } n+2r = (n+r) + r \\
 &= \sum_{r=1}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n-1+2r} \\
 &\quad + \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= J_{n-1} + \sum_{r=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+1-2j} \\
 &\quad \text{taking } j=r-1 \\
 &= J_{n-1} - J_{n+1}. \text{ This proves the result.}
 \end{aligned}$$

IV. $2nJ_n = x(J_{n-1} + J_{n+1})$.

[Agra 72, 56 ; Final 66 ; Raj. 59]

This result is also directly obtained by subtracting formula I and II. A direct proof is given below.

We can write

$$\begin{aligned}
 2nJ_n &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} (2n) \left(\frac{x}{2}\right)^{n+2r} \\
 &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} (2n+2r-2r) \left(\frac{x}{2}\right)^{n+2r} \\
 &\quad \text{taking } 2n = 2n+2r-2r \\
 &= x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &\quad - \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} x \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= xJ_{n-1} - x \sum_{r=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+2j+1} \\
 &\quad \text{taking } j=r-1 \\
 &= xJ_{n-1} + xJ_{n+1}.
 \end{aligned}$$

Thus $2nJ_n = x[J_{n-1} + J_{n+1}]$.

V. $\frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1}$.

Recurrence formula I as obtained above is

$$xJ'_n = nJ_n - xJ_{n+1}, \text{ or } xJ'_n - nJ_n = -xJ_{n+1}.$$

Multiplying it by x^{-n-1} , we get

$$x^{-n} J'_n - nx^{-n-1} J_n = -x^{-n} J_{n+1},$$

This can be written as

8.4. Recurrence formulae

1. $xJ'_n = nJ_n - xJ_{n+1}$.

By definition,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

Differentiating it with respect to x , we get

$$J'_n = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \cdot \frac{1}{2}$$

$$\begin{aligned} \therefore xJ'_n &= n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad + x \sum_{r=0}^{\infty} \frac{(-1)^r \cdot 2r}{2 \cdot (r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= nJ_n + x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \end{aligned}$$

in the second summation r takes values from 1 to ∞

$$\begin{aligned} &= nJ_n + x \sum_{j=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+j+2)} \left(\frac{x}{2}\right)^{n+2j+1} \\ &\quad \text{taking } j=r-1 \\ &= nJ_n - x \sum_{j=0}^{\infty} \frac{(-1)^j}{(j)! \Gamma\{(n+1)+j+1\}} \left(\frac{x}{2}\right)^{n+1+2j} \end{aligned}$$

Thus $xJ'_n = nJ_n - xJ_{n+1}$.

II. $xJ'_n = -nJ_n + xJ_{n-1}$.

[Vikram 63 ; Agra 54]

As above, we have

$$\begin{aligned} xJ'_n(x) &= \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &= \sum_{r=0}^{\infty} \frac{(-1)^r \{(2n+2r)-n\}}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad \text{as } n+2r = (2n+2r) - n \\ &= -n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad + x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= -nJ_n + x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma\{(n-1)+r+1\}} \left(\frac{x}{2}\right)^{n-1+2r} \end{aligned}$$

or $xJ'_n = -nJ_n + xJ_{n-1}$.

III. $2J'_n = J_{n-1} - J_{n+1}$.

[Vikram 63 ; Agra 63 ; 58, 56, 52 , Final 66]

This is obtained directly by adding the recurrence relations I and II. We give an alternative proof as follows :

$$J_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

8.4. Recurrence formulae

1. $xJ'_n = nJ_n - xJ_{n+1}$.

By definition,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

Differentiating it with respect to x , we get

$$J'_n = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \cdot \frac{1}{2}$$

$$\begin{aligned} \therefore xJ'_n &= n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad + x \sum_{r=0}^{\infty} \frac{(-1)^r \cdot 2r}{2 \cdot (r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= nJ_n + x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \end{aligned}$$

in the second summation r takes values from 1 to ∞

$$\begin{aligned} &= nJ_n + x \sum_{j=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+j+2)} \left(\frac{x}{2}\right)^{n+2j+1} \\ &\quad \text{taking } j=r-1 \\ &= nJ_n - x \sum_{j=0}^{\infty} \frac{(-1)^j}{(j)! \Gamma\{(n+1)+j+1\}} \left(\frac{x}{2}\right)^{n+1+2j} \end{aligned}$$

Thus $xJ'_n = nJ_n - xJ_{n+1}$.

II. $xJ'_n = -nJ_n + xJ_{n-1}$.

[Vikram 63 ; Agra 54]

As above, we have

$$\begin{aligned} xJ'_n(x) &= \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &= \sum_{r=0}^{\infty} \frac{(-1)^r \{(2n+2r)-n\}}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad \text{as } n+2r = (2n+2r) - n \\ &= -n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad + x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= -nJ_n + x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma\{(n-1)+r+1\}} \left(\frac{x}{2}\right)^{n-1+2r} \end{aligned}$$

or $xJ'_n = -nJ_n + xJ_{n-1}$.

III. $2J'_n = J_{n-1} - J_{n+1}$.

[Vikram 63 ; Agra 63 ; 58, 56, 52 , Final 66]

This is obtained directly by adding the recurrence relations I and II. We give an alternative proof as follows :

$$J_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

$$\frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1},$$

which is the required result.

Alternatively, we have

$$\begin{aligned} \frac{d}{dx} (x^{-n} J_n) &= \frac{d}{dx} \left[x^{-n} \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right] \\ &= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} 2^{-n} \left(\frac{x}{2}\right)^{2r} \right] \\ &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} 2^{-n} \left(\frac{x}{2}\right)^{2r-1} \\ &= x^{-n} \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= x^{-n} \sum_{j=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+1+2j} \\ &\quad \text{taking } j=r-1 \\ &= -x^{-n} \sum_{j=1}^{\infty} \frac{(-1)^j}{(j)! \Gamma\{(n+1)+j+1\}} \left(\frac{x}{2}\right)^{n+1+2j} \\ &= -x^{-n} J_{n+1}. \end{aligned}$$

$$\text{VI. } \frac{d}{dx} (x^n J_n) = x^n J_{n-1}.$$

[Vikram 62 ; Agra 72, 66, 57 ; Raj. 59]

Recurrence formula II is $xJ'_n = -nJ_n + xJ_{n-1}$

or $xJ'_n + nJ_n = xJ_{n-1}$.

Multiplying it by x^{n-1} , we get

$$x^n J'_n + n x^{n-1} J_n = x^n J_{n-1}.$$

This can be written as

$$\frac{d}{dx} (x^n J_n) = x^n J_{n-1}$$

which is the required result.

Alternatively, we have

$$\begin{aligned} \frac{d}{dx} (x^n J_n) &= \left[x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right] \\ &= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} 2^n \left(\frac{x}{2}\right)^{2n+2r} \right] \\ &= 2^n \sum_{r=0}^{\infty} \frac{(-1)^r (n+r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{2n+2r-1} \\ &= x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= x^n J_{n-1}. \end{aligned}$$

8.4. A generating function

To show that when n is a positive integer $J_n(x)$ is the coefficient of t^n in the expansion of $e^{\frac{1}{2}x(t-1/t)}$ in ascending powers of t .

[Vikram 62 ; Raj. 61 ; Agra 72, 70, 66; 63, 62]

We have $e^{xt} = \sum_{r=0}^{\infty} \frac{1}{(r)!} \left(\frac{xt}{2}\right)^r$

and $e^{-x/2t} = \sum_{s=0}^{\infty} \frac{1}{(s)!} \left(\frac{-x}{2t}\right)^s$

$$\begin{aligned} \text{Therefore } e^{\frac{1}{2}x(t-1/t)} &= \sum_{r=0}^{\infty} \frac{1}{(r)!} \left(\frac{xt}{2}\right)^r \sum_{s=0}^{\infty} \frac{1}{(s)!} \left(-\frac{x}{t}\right)^s \\ &= \sum_{r=0}^{\infty} \sum_{s=0}^{\infty} \frac{(-1)^s}{(r)! (s)!} \left(\frac{x}{2}\right)^{r+s} t^{r-s}. \end{aligned} \quad \dots(1)$$

We want to collect coefficient of t^n from the right, for that suppose that $n=r-s$, i.e., $r=n+s$. Now give values to s from 0 to infinity, while r takes values such that $r=n+s$.

Thus the coefficient of t^n on the right of (1) is

$$\begin{aligned} &\sum_{s=0}^{\infty} \frac{(-1)^s}{(n+s)! (s)!} \left(\frac{x}{2}\right)^{n+2s} \\ &= \sum_{s=0}^{\infty} \frac{(-1)^s}{(s)! \Gamma(n+s+1)} \left(\frac{x}{2}\right)^{n+2s} \end{aligned}$$

as $(n+s)! = \Gamma(n+s+1)$

$= J_n(x).$

Again to collect the coefficient of t^{-n} from the right of (1), give values to r from 0 to infinity while s take values such that $s=n+r$.

In this way coefficient of t^{-n} on the right of (1)

$$\begin{aligned} &= \sum_{r=0}^{\infty} \frac{(-1)^{n+r}}{(r)! (n+r)!} \left(\frac{x}{2}\right)^{n+2r} \\ &= (-1)^n J_n(x) \\ &= J_{-n}(x) \text{ as } J_{-n}(x) = (-1)^n J_n(x). \end{aligned}$$

$$\therefore e^{\frac{1}{2}x(t-1/t)} = \sum_{n=-\infty}^{\infty} t^n J_n(x).$$

8.5. Trigonometric expansions

To show that

$$\cos(x \sin \phi) = J_0(x) - 2 [\cos 2\phi J_2(x) + \cos 4\phi J_4(x) + \dots],$$

$$\sin(x \sin \phi) = 2 [\sin \phi J_1(x) + \sin 3\phi J_3(x) + \dots],$$

$$\cos(x \cos \phi) = J_0(x) - 2 [\cos 2\phi J_2(x) - \cos 4\phi J_4(x) + \dots],$$

$$\sin(x \cos \phi) = 2 [\cos \phi J_1(x) - \cos 3\phi J_3(x) + \dots].$$

Proof. We have

$$e^{\frac{1}{2}x(t-1/t)} = \sum_{n=-\infty}^{\infty} t^n J_n(x).$$

Putting $t = e^{i\phi}$, so that $\frac{1}{2}(t-1/t) = i \sin \phi$, we get

8.4. Recurrence formulae

1. $xJ'_n = nJ_n - xJ_{n+1}$.

By definition,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

Differentiating it with respect to x , we get

$$J'_n = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \cdot \frac{1}{2}$$

$$\therefore xJ'_n = n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

$$+ x \sum_{r=0}^{\infty} \frac{(-1)^r \cdot 2r}{2 \cdot (r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$= nJ_n + x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1}$$

in the second summation r takes values from 1 to ∞

$$= nJ_n + x \sum_{j=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+j+2)} \left(\frac{x}{2}\right)^{n+2j+1}$$

taking $j=r-1$

$$= nJ_n - x \sum_{j=0}^{\infty} \frac{(-1)^j}{(j)! \Gamma\{(n+1)+j+1\}} \left(\frac{x}{2}\right)^{n+1+2j}$$

Thus $xJ'_n = nJ_n - xJ_{n+1}$.

II. $xJ'_n = -nJ_n + xJ_{n-1}$.

[Vikram 63 ; Agra 54]

As above, we have

$$xJ'_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^r \{(2n+2r)-n\}}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

as $n+2r = (2n+2r) - n$

$$= -n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

$$+ x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$= -nJ_n + x \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma\{(n-1)+r+1\}} \left(\frac{x}{2}\right)^{n-1+2r}$$

or $xJ'_n = -nJ_n + xJ_{n-1}$.

III. $2J'_n = J_{n-1} - J_{n+1}$.

[Vikram 63 ; Agra 63 ; 58, 56, 52 , Final 66]

This is obtained directly by adding the recurrence relations I and II. We give an alternative proof as follows :

$$J_n = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}
 2J'_n &= \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= \sum_{r=0}^{\infty} \frac{(-1)^r \{(n+r)+r\}}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &\quad \text{taking } n+2r = (n+r) + r \\
 &= \sum_{r=1}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n-1+2r} \\
 &\quad + \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= J_{n-1} + \sum_{r=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+1-2j} \\
 &\quad \text{taking } j=r-1 \\
 &= J_{n-1} - J_{n+1}, \text{ This proves the result.}
 \end{aligned}$$

IV. $2nJ_n = x(J_{n-1} + J_{n+1})$.

[Agra 72, 56 ; Final 66 ; Raj. 59]

This result is also directly obtained by subtracting formula I and II. A direct proof is given below.

We can write

$$\begin{aligned}
 2nJ_n &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} (2n) \left(\frac{x}{2}\right)^{n+2r} \\
 &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} (2n+2r-2r) \left(\frac{x}{2}\right)^{n+2r} \\
 &\quad \text{taking } 2n = 2n+2r-2r \\
 &= x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &\quad - \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} x \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= xJ_{n-1} - x \sum_{r=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+2j+1} \\
 &\quad \text{taking } j=r-1 \\
 &= xJ_{n-1} + xJ_{n+1}.
 \end{aligned}$$

Thus $2nJ_n = x(J_{n-1} + J_{n+1})$.

V. $\frac{d}{dx}(x^{-n}J_n) = -x^{-n}J_{n+1}$.

Recurrence formula I as obtained above is

$$xJ'_n = nJ_n - xJ_{n+1}, \text{ or } xJ'_n - nJ_n = -xJ_{n+1}.$$

Multiplying it by x^{-n-1} , we get

$$x^{-n}J'_n - nx^{-n-1}J_n = -x^{-n}J_{n+1},$$

This can be written as

8.4. A generating function

To show that when n is a positive integer $J_n(x)$ is the coefficient of t^n in the expansion of $e^{\frac{1}{2}x(t-1/t)}$ in ascending powers of t .

[Vikram 62 ; Raj. 61 ; Agra 72, 70, 66; 63, 62]

We have $e^{\frac{1}{2}xt} = \sum_{r=0}^{\infty} \frac{1}{(r)!} \left(\frac{xt}{2}\right)^r$

and $e^{-\frac{1}{2}x/t} = \sum_{s=0}^{\infty} \frac{1}{(s)!} \left(\frac{-x}{2t}\right)^s$

$$\begin{aligned} \text{Therefore } e^{\frac{1}{2}x(t-1/t)} &= \sum_{r=0}^{\infty} \frac{1}{(r)!} \left(\frac{xt}{2}\right)^r \sum_{s=0}^{\infty} \frac{1}{(s)!} \left(\frac{-x}{2t}\right)^s \\ &= \sum_{r=0}^{\infty} \sum_{s=0}^{\infty} \frac{(-1)^s}{(r)! (s)!} \left(\frac{x}{2}\right)^{r+s} t^{r-s}. \end{aligned} \quad \dots(1)$$

We want to collect coefficient of t^n from the right, for that suppose that $n=r-s$, i.e., $r=n+s$. Now give values to s from 0 to infinity, while r takes values such that $r=n+s$.

Thus the coefficient of t^n on the right of (1) is

$$\begin{aligned} &\sum_{s=0}^{\infty} \frac{(-1)^s}{(n+s)! (s)!} \left(\frac{x}{2}\right)^{n+2s} \\ &= \sum_{s=0}^{\infty} \frac{(-1)^s}{(s)! \Gamma(n+s+1)} \left(\frac{x}{2}\right)^{n+2s} \end{aligned}$$

as $(n+s)! = \Gamma(n+s+1)$

$= J_n(x).$

Again to collect the coefficient of t^{-n} from the right of (1), give values to r from 0 to infinity while s take values such that $s=n+r$.

In this way coefficient of t^{-n} on the right of (1)

$$\begin{aligned} &= \sum_{r=0}^{\infty} \frac{(-1)^{n+r}}{(r)! (n+r)!} \left(\frac{x}{2}\right)^{n+2r} \\ &= (-1)^n J_n(x) \\ &= J_{-n}(x) \text{ as } J_{-n}(x) = (-1)^n J_n(x). \end{aligned}$$

$$\therefore e^{\frac{1}{2}x(t-1/t)} = \sum_{n=-\infty}^{\infty} t^n J_n(x).$$

8.5. Trigonometric expansions

To show that

$$\cos(x \sin \phi) = J_0(x) + 2 [\cos 2\phi J_2(x) + \cos 4\phi J_4(x) + \dots],$$

$$\sin(x \sin \phi) = 2 [\sin \phi J_1(x) + \sin 3\phi J_3(x) + \dots],$$

$$\cos(x \cos \phi) = J_0(x) - 2 [\cos 2\phi J_2(x) - \cos 4\phi J_4(x) + \dots],$$

$$\sin(x \cos \phi) = 2 [\cos \phi J_1(x) - \cos 3\phi J_3(x) + \dots].$$

Proof. We have

$$e^{\frac{1}{2}x(t-1/t)} = \sum_{n=-\infty}^{\infty} t^n J_n(x).$$

Putting $t = e^{i\phi}$, so that $\frac{1}{2}(t-1/t) = i \sin \phi$. we get

$$\frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1},$$

which is the required result.

Alternatively, we have

$$\begin{aligned} \frac{d}{dx} (x^{-n} J_n) &= \frac{d}{dx} \left[x^{-n} \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right] \\ &= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} 2^{-n} \left(\frac{x}{2}\right)^{2r} \right] \\ &= \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} 2^{-n} \left(\frac{x}{2}\right)^{2r-1} \\ &= x^{-n} \sum_{r=0}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= x^{-n} \sum_{j=0}^{\infty} \frac{(-1)^{j+1}}{(j)! \Gamma(n+1+j+1)} \left(\frac{x}{2}\right)^{n+1+2j} \\ &\quad \text{taking } j=r-1 \\ &= -x^{-n} \sum_{j=1}^{\infty} \frac{(-1)^j}{(j)! \Gamma\{(n+1)+j+1\}} \left(\frac{x}{2}\right)^{n+1+2j} \\ &= -x^{-n} J_{n+1}. \end{aligned}$$

$$\text{VI. } \frac{d}{dx} (x^n J_n) = x^n J_{n-1}.$$

[Vikram 62 ; Agra 72, 66, 57 ; Raj. 59]

Recurrence formula II is $xJ'_n = -nJ_n + xJ_{n-1}$

or $xJ'_n + nJ_n = xJ_{n-1}$.

Multiplying it by x^{n-1} , we get

$$x^n J'_n + n x^{n-1} J_n = x^n J_{n-1}.$$

This can be written as

$$\frac{d}{dx} (x^n J_n) = x^n J_{n-1}$$

which is the required result.

Alternatively, we have

$$\begin{aligned} \frac{d}{dx} (x^n J_n) &= \left[x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right] \\ &= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r+1)} \cdot 2^n \left(\frac{x}{2}\right)^{2n+2r} \right] \\ &= 2^n \sum_{r=0}^{\infty} \frac{(-1)^r (n+r)}{(r)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{2n+2r-1} \\ &= x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{(r)! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= x^n J_{n-1}. \end{aligned}$$

Solution. We have by definition

$$J_0(x) = \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{2r}$$

Differentiating it w.r.t. x , we get

$$\frac{d}{dx} J_0(x)$$

$$= \sum_{r=1}^{\infty} (-1)^r \frac{1}{(r)! (r-1)!} \left(\frac{x}{2}\right)^{2r-1}$$

$$= \sum_{j=0}^{\infty} (-1)^{j+1} \frac{1}{(j+1)! (j)!} \left(\frac{x}{2}\right)^{2j+1}$$

where $j = r-1$

$$= - \sum_{j=0}^{\infty} (-1)^j \frac{1}{(j)! (j+1)!} \left(\frac{x}{2}\right)^{2j+1} = -J_1(x).$$

8.3. When n is positive integer,

$$J_{-n}(x) = (-1)^n J_n(x).$$

Proof. We have

$$J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{(r)! \Gamma(r-n+1)} \left(\frac{x}{2}\right)^{-n+2r}$$

In this series Γ of a negative number occurs in denominator, for terms when $r=0, 1, 2, \dots, (n-1)$ and Γ of a negative number being infinite, all these terms are zero.

Thus terms are obtained from $r=n$ onwards

$$\text{or } J_{-n}(x) = \sum_{r=n}^{\infty} (-1)^r \frac{1}{(r)! \Gamma(r-n+1)} \left(\frac{x}{2}\right)^{-n+2r}$$

Now take $r=n+j$, so that when $r=n$, $j=0$ and

$$J_{-n}(x) = \sum_{j=0}^{\infty} (-1)^{j+n} \frac{1}{(n+j)! \Gamma(n+j-n+1)} \left(\frac{x}{2}\right)^{-n+2n+2j}$$

$$= \sum_{j=0}^{\infty} (-1)^{j+n} \frac{1}{(n+j)! \Gamma(j+1)} \left(\frac{x}{2}\right)^{n+2j}$$

$$= (-1)^n \sum_{j=0}^{\infty} (-1)^j \frac{1}{(n+j+1) \cdot (j)!} \left(\frac{x}{2}\right)^{n+2j}$$

as $(n+j)! = \Gamma(n+j+1)$ and $\Gamma(j+1) = (j)!$

$$= (-1)^n J_n(x).$$

Note. Because of this relation in $J_n(x)$ and $J_{-n}(x)$, the two functions are not independent when n is an integer. Therefore in this case a second solution of Bessel's equation is to be found.

Ex. State the Bessel's differetial equation and prove that its most general solution is $y = AJ_n(x) + BJ_{-n}(x)$ where A and B are arbitrary constants. [Agra 1973]